Josephine Adelegan

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EDUCATION

Howard University **Washington D.C.**
Bachelor of Science, Chemical Engineering *Graduation: May 2022*
Magna Cum Laude

Howard University **Washington D.C.**
PhD Student, Civil and Environmental Engineering *January 2024 – December 2027*

- **Scholarships & Honors:**
  - *AIAS CRIT Scholar, American Institute of Architecture Students (AIAS)* *August 2025 – May 2026*
  - *Ernest E. Just-Percy L. Julian Graduate Research Fellow* *August 2024 – May 2025*

EXPERIENCE

Graduate Research with Dr. Kimberly Jones **Washington D.C.**
Graduate Researcher - PhD *January 2024 - Present*

- Conducting predictive modeling of membrane fouling in radioactive wastewater using Artificial Neural Networks (ANN) and Genetic Programming (GP) with data from DOE sites.
- Designing experimental validation protocols, including SEM and EDS analysis, to assess fouling layers and test sustainable cleaning strategies like TiO_2 coatings and ultrasound.
- Aligning research with global clean water access goals by exploring applications of nuclear wastewater findings in low-resource and developing

DC (District of Columbia) Water **Washington D.C.**
Research & Development Operations Intern *May 2024 - June 2024*

- Analyzed polymer performance by evaluating the effectiveness of plant-prepared and lab-prepared polymers through measurements of viscosity, charge density, and solid content to optimize sludge dewatering processes.
- Implemented monitoring systems by setting up and managing GoPro cameras and sensors on the Belt Filter Press (BFP) to capture data for real-time polymer dosing adjustments.
- Enhanced sludge conditioning by developing survey tools and analyzing visual data to refine polymer dosing strategies, ensuring precise mixing and improving dewatering efficiency.

D3 Systems *Government Subcontract supporting USAID* **Tysons, Virginia**
Third-Party Monitoring Project Support *May 2023 - December 2023*

- Managed and completed all third-party monitoring projects for a variety of clients, including site activities in Afghanistan across a variety of aid topics.
- Coordinated completion of approximately 200 site visits per month in communication with the project lead.

Standard Solar **Rockville, Maryland**
Engineering (Photovoltaic) Intern *May 2023*

- Learned to use AutoCAD, PVsyst, HelioScope, and conducted detailed energy modeling calculations, including string sizing, inverter matching, and voltage drop analysis.
- Provided conceptual design deliverables to support commercial and/or utility-scale solar electric project development.
- Executed engineering and technical tasks of assigned projects primarily regarding early-stage evaluation, conceptual design development and sensitivity analysis of design variables for optimization.

AstraZeneca **Gaithersburg, Maryland**
Production Technician *September 2022 - March 2023*

- Executed all steps of upstream, downstream, and central services operations in biopharmaceutical manufacturing.
- Generated and regularly updated SOPs, MPRs, and SPRs for equipment and procedures utilized in manufacturing of biopharma products.

Global Network for Environment and Economic Development Research, *African Environment and Sustainable Development Institution* **Ibadan, Nigeria**
Environmental Engineering Intern on the following food-water-energy nexus innovation *June 2021 - July 2021*

- ***Power to the Poor***: *Conversion of Cassava Wastes into Off-grid Lighting and Bio-Fertilizer* in Nigeria. The Innovation is funded by SEED International Awards including UNEP, Swiss Re-Insurance, GPPi, ESKOM, the Government of United Kingdom, Germany, Netherlands, Norway and United States.

Undergraduate Research with Dr. Preethi L. Chandran **Washington D.C. (Remote)**
Undergraduate Researcher *June 2020 - August 2020*

- Utilized MATLAB to plot 3D representations of biomaterials.
- Conducted research and analyzed structural properties, mechanics, and potential impact of spiderwebs on biotechnology.

Global Network for Environment and Economic Development Research, *African Environment and Sustainable Development Institution* **Ibadan, Nigeria**
Environmental Engineering Intern on the following food-water-energy nexus innovation *June 2019 - July 2019*

- ***Cows to Kilowatts***: *Anaerobic Bio-digestion of Abattoir Wastes* Converted into Bioenergy and Bio-Fertilizer in Africa (Nigeria and Kenya). The innovation was funded by the ***United Nations Development Program***

United Nations University *Academic and research arm of the United Nations* **Accra, Ghana**
Environmental Engineering Intern *July 2018 - August 2018*

- Explored topics in process design, power supply maintenance, and equipment efficiency and sustainability.
- Progressed global think tank's thought leadership on sustainability and the water industry in West Africa by presenting research findings on a solar energy and turbine electric dam solution, which reached over 150 stakeholders, including the General Manager and Consultants.
- Applied Gibbs Free Energy Theory to assess system thermodynamics, increasing plant efficiency by 16%.

PROJECTS

Thermal & Envelope Engineer | U.S. Department of Energy Solar Decathlon – Team Resonance **Washington D.C.**
Howard University *January 2025 – Present*

Grace Farms Foundation Design Competition *Spring 2025*

Capstone Design Project *January 2022 - May 2022*

Fundamentals to Environmental Engineering Design Project *January 2022 - May 2022*

CERTIFICATIONS

Design Justice Badge | **Drexel University & Howard University (2024)**
- Recognized for inclusive design addressing climate justice, disability access, and sustainability.

Clean Energy Justice Innovator Badge | **Drexel University & Howard University (2025)**
- Recognized for advancing clean energy equity through community-centered education, innovation, and collaboration.

TECHNICAL SKILLS	Microsoft Office Suite (Excel and PowerPoint Advanced), Google Applications, DWSIM (The Open-Source Chemical Process Simulator), Economic Analysis, Stress Test, Sensitivity Analysis, pH conductivity, wet chemistry, cGMP, SOPs, MPR, and SPR proficiency, AutoCAD, PVSyst, Helioscope.
SOFT SKILLS	Travel flexibility, clear communication across all levels, collaborative problem solving, creative mindset, people-oriented, and innovative thinking.
LANGUAGES	MATLAB programming language, R programming language (Beginner), Yoruba language (Native), French language (Intermediate).
RELEVANT COURSEWORK	Engineering Design, Thermodynamics I & II, Transport Phenomena I & II, Mass Transfer, Numerical Methods, Separation Process, Chemical Engineering Lab, Fluid Mechanics, Chemical Engineering Kinetics, Process Control, Fundamentals to Environmental Engineering, Frontiers in Nanoscience Engineering Technology, Unit Operations, Environmental Chemistry, Advanced Transport, Advanced Hydrology, Environmental Microbiology.

PUBLICATIONS & REPORTS

1. **Adelegan, J.** (Presenting Author, Graduate Student), Bhattarai, Y. (Graduate Student), Sharma, S., Jones, K. (Faculty Advisors), *"Innovative Predictive Modeling and Experimental Approaches to Mitigate Membrane Fouling in Nuclear Wastewater Treatment"*, **Waste Management Symposium 2026**, Phoenix Convention Center, Phoenix, AZ, United States. March 11, 2026.

2. **Adelegan, J.** (Presenting Author, Graduate Student), Bhattarai, Y., Sharma, S., Jones, K. (Faculty Advisors), *"Data-Driven Strategies for Predicting and Mitigating Membrane Fouling in High-Risk Nuclear Wastewater Systems"*, **Waste Management Symposium 2026**, Phoenix Convention Center, Phoenix, AZ, United States. March 9, 2026.

3. **Adelegan, J.** (Presenting Author, Graduate Student), *"Predictive Modeling and Bench-Scale Analysis of Membrane Fouling for Sustainable Water Management"*, **114th Annual Meeting of the Association of Collegiate Schools of Architecture (ACSA)**, Loews Hotel, Chicago, IL, United States. March 27, 2026.

LEADERSHIP

CRESP (Consortium For Risk Evaluation with Stakeholder Participation) Early Career Program **Washington, DC**
Field Visit Chair *August 2025–Present*

Crisis Text Line **New York, NY**
Volunteer Crisis Counselor *December 2021–December 2023*

Girl Scouts of Nassau County **East Hampton, NY**
Unit Leader *July 2021–August 2021*